DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE; THE FACILITY OWNER'S QUALIFIED PROCEDURES AND APPLICABLE CODES GOVERN.

WO-400-01 — Overhaul pump P-101A (mechanical seal and bearings)

Area	A400 Pumps	**Task**	PO Pump overhaul
Priority	3 (reliability)	**Issued by**	Facility owner (demonstration)
Primary asset	**P-101A** — P-101A centrifugal pump — E 154'-0" N -18'-6" EL 100'-9"		
Equipment	P-101A	**Lines**	6"-P150-P-1001, 4"-P300-P-1003, 3/4"-P150-W-3017
Isometrics	ISO-6"-P150-P-1001, ISO-4"-P300-P-1003, ISO-3/4"-P150-W-3017 (generated); owner's drawings ISO-P-1001, ISO-P-1003, ISO-W-3017		
Permits	General work permit, Lifting plan		

Reason

Mechanical seal leaking; overhaul seal and bearings.

Scope of work

1. Isolate P-101A electrically and per the isolation list; drain the casing.
2. Record as-found shaft alignment; remove the coupling guard and spacer.
3. Break the suction and discharge nozzle joints; remove the pump to the workshop.
4. Replace the mechanical seal and the bearings; inspect the impeller and wear rings.
5. Reinstall; new nozzle gaskets and studs tightened per SW-WO-5; align to the owner's alignment criteria.
6. Remove blinds and isolations; run test.

Affected items: P-101A

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-NDE-5, SW-TEST-2, SW-WO-1, SW-WO-2, SW-WO-3, SW-WO-5, SW-WO-7

Owner's isometric drawings (SW-IMP-6)

Drawing	Line drawn	Work line	Sheets	Match to the plant model
ISO-P-1001	6"-P150-P-1001	6"-P150-P-1001	5	1 error(s), 4 warning(s) — see below
ISO-P-1003	4"-P300-P-1003	4"-P300-P-1003	3	3 error(s), 3 warning(s) — see below
ISO-W-3017	3/4"-P150-W-3017	3/4"-P150-W-3017	1	agrees with the model

ISO-P-1001 (PDF) · ISO-P-1003 (PDF) · ISO-W-3017 (PDF) — the generated isometrics follow the ITP.

Drawing	Sheet	Level	Finding
ISO-P-1001	5	error	sheet 5: CONT'D ON ISO-P-1002 prints E 154'-5" N -12'-6" EL 102'-0" for E 154'-5" N -11'-6" EL 102'-0": the minus sign applies to the feet only; read as written the point is 12.00 in away and 12.00 in off where 6"-P150-P-1001 meets 6"-P150-P-1002 (E 154'-5" N -11'-6" EL 102'-0")
ISO-P-1001	5	warning	W21 (6"-P150-P-1001/W21, sheet 5) is labelled again on sheet 4; it is 58.0 in from the nearest break
ISO-P-1001	4	warning	branch weld 3/4"-P300-I-7030/W1 (E 151'-6" N 0" EL 117'-5", sheet 4) has no label
ISO-P-1001	4	warning	branch weld 3/4"-P300-I-7001/W1 (E 152'-6" N 0" EL 117'-5", sheet 4) has no label
ISO-P-1001	5	warning	TO label 'P-101A NOZ. S' is not on the drawing
ISO-P-1003	1	error	sheet 1: CONT'D ON ISO-P-1004 prints E 154'-3 5/16" N -17'-6" EL 107'-6" for E 154'-3 5/16" N -16'-6" EL 107'-6": the minus sign applies to the feet only; read as written the point is 12.00 in away and 12.00 in off where 4"-P300-P-1003 meets 4"-P300-P-1004 (E 154'-3 5/16" N -16'-6" EL 107'-6")
ISO-P-1003	2	error	sheet 2: CONT'D ON ISO-P-1003 SH 3 prints E 110'-6" N -4'-9" EL 118'-6" for E 110'-6" N -3'-3" EL 118'-6": the minus sign applies to the feet only; read as written the point is 18.00 in away (still on the centreline, at the wrong place along it)
ISO-P-1003	3	error	sheet 3: CONT'D FROM ISO-P-1003 SH 2 prints E 110'-6" N -4'-9" EL 118'-6" for E 110'-6" N -3'-3" EL 118'-6": the minus sign applies to the feet only; read as written the point is 18.00 in away (still on the centreline, at the wrong place along it)
ISO-P-1003	3	warning	W18 (4"-P300-P-1003/W18, sheet 3) is labelled again on sheet 2; it is 5.7 in from the nearest break
ISO-P-1003	2	warning	branch weld 3/4"-P300-I-7003/W1 (E 110'-6" N -2'-6" EL 118'-9 5/16", sheet 2) has no label
ISO-P-1003	3	warning	branch weld 3/4"-P300-SP-7022/W1 (E 110'-6" N -4'-0" EL 118'-9 5/16", sheet 3) has no label

Assets in the plant model (SW-WO-7)

Tag	Kind	Description	Line	Location
P-101A	equipment	P-101A centrifugal pump		E 154'-0" N -18'-6" EL 100'-9"
P-101A-M	motor	Motor of P-101A		E 154'-0" N -18'-6" EL 100'-9"
6"-P150-P-1001/B7	item	VALVE, GATE, WEDGE, OS&Y 6" P150	6"-P150-P-1001	E 154'-0" N -13'-5-1/8" EL 102'-0"
6"-P150-P-1001/B5	item	FLANGE WELD-NECK 6" P150 RAISED FACE	6"-P150-P-1001	E 154'-0" N -14'-2-1/4" EL 102'-0"
4"-P300-P-1003/A5	item	VALVE, GATE, WEDGE, OS&Y 3" P300	4"-P300-P-1003	E 154'-0" N -16'-6" EL 105'-2-3/16"
4"-P300-P-1003/A2	item	VALVE, SWING CHECK 3" P300	4"-P300-P-1003	E 154'-0" N -16'-6" EL 104'-1/8"
3/4"-P150-W-3017/A3	item	VALVE, BALL, FLOATING 3/4" P150	3/4"-P150-W-3017	E 156'-0" N -4'-0" EL 123'-7-11/16"
3/4"-P150-W-3017/A6	item	FLANGE WELD-NECK 3/4" P150 RAISED FACE	3/4"-P150-W-3017	E 156'-0" N -4'-0" EL 124'-4-5/8"

Isolation list (SW-WO-3)

#	Kind	Tag	Line	Action	Location	Fitted ✓	Removed ✓
1	electrical	P-101A-M		Motor of P-101A: isolated, locked and tried at the starter			
2	valve	6"-P150-P-1001/B7	6"-P150-P-1001	Close and lock VALVE, GATE, WEDGE, OS&Y 6" P150 (6"-P150-P-1001/B7) on P-101A-S	E 154'-0" N -13'-5-1/8" EL 102'-0"		
3	blind	6"-P150-P-1001/B5	6"-P150-P-1001	Fit a blind between 6"-P150-P-1001/B7 and 6"-P150-P-1001/B5, P-101A side	E 154'-0" N -14'-4-3/16" EL 102'-0"		
4	valve	4"-P300-P-1003/A5	4"-P300-P-1003	Close and lock VALVE, GATE, WEDGE, OS&Y 3" P300 (4"-P300-P-1003/A5) on P-101A-D	E 154'-0" N -16'-6" EL 105'-2-3/16"		
5	blind	4"-P300-P-1003/A2	4"-P300-P-1003	Fit a blind between 4"-P300-P-1003/A5 and 4"-P300-P-1003/A2, P-101A side	E 154'-0" N -16'-6" EL 103'-5"		
6	valve	3/4"-P150-W-3017/A3	3/4"-P150-W-3017	Close and lock VALVE, BALL, FLOATING 3/4" P150 (3/4"-P150-W-3017/A3) on P-101A-F	E 156'-0" N -4'-0" EL 123'-7-11/16"		
7	blind	3/4"-P150-W-3017/A6	3/4"-P150-W-3017	Fit a blind between 3/4"-P150-W-3017/A3 and 3/4"-P150-W-3017/A6, P-101A side	E 156'-0" N -4'-0" EL 124'-2-3/8"		

Parts

Description	Qty	Replaces
GASKET 4" P150 COMPRESSED SHEET	1	6"-P150-P-1001/B1
STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P150	1	6"-P150-P-1001/B3
GASKET 3" P300 COMPRESSED SHEET	1	4"-P300-P-1003/A1
STUD SET 4 × 0.875" × 5-3/4" WITH 2 NUTS, FOR 3" P300	1	4"-P300-P-1003/A3
Mechanical seal kit for P-101A	1	
Bearing set for P-101A	1	

Bolt-up (SW-WO-5)

Joint	Studs	Preload per stud (lbf)	Passes	Done ✓
6"-P150-P-1001/B1	4 × 1" studs (6"-P150-P-1001/B3)	35,343	3	
4"-P300-P-1003/A1	4 × 0.875" studs (4"-P300-P-1003/A3)	27,059	3	

Test — Run test

Description	Running at duty: no seal leak; bearing temperature and vibration within the owner's limits
Basis	SW-TEST-2

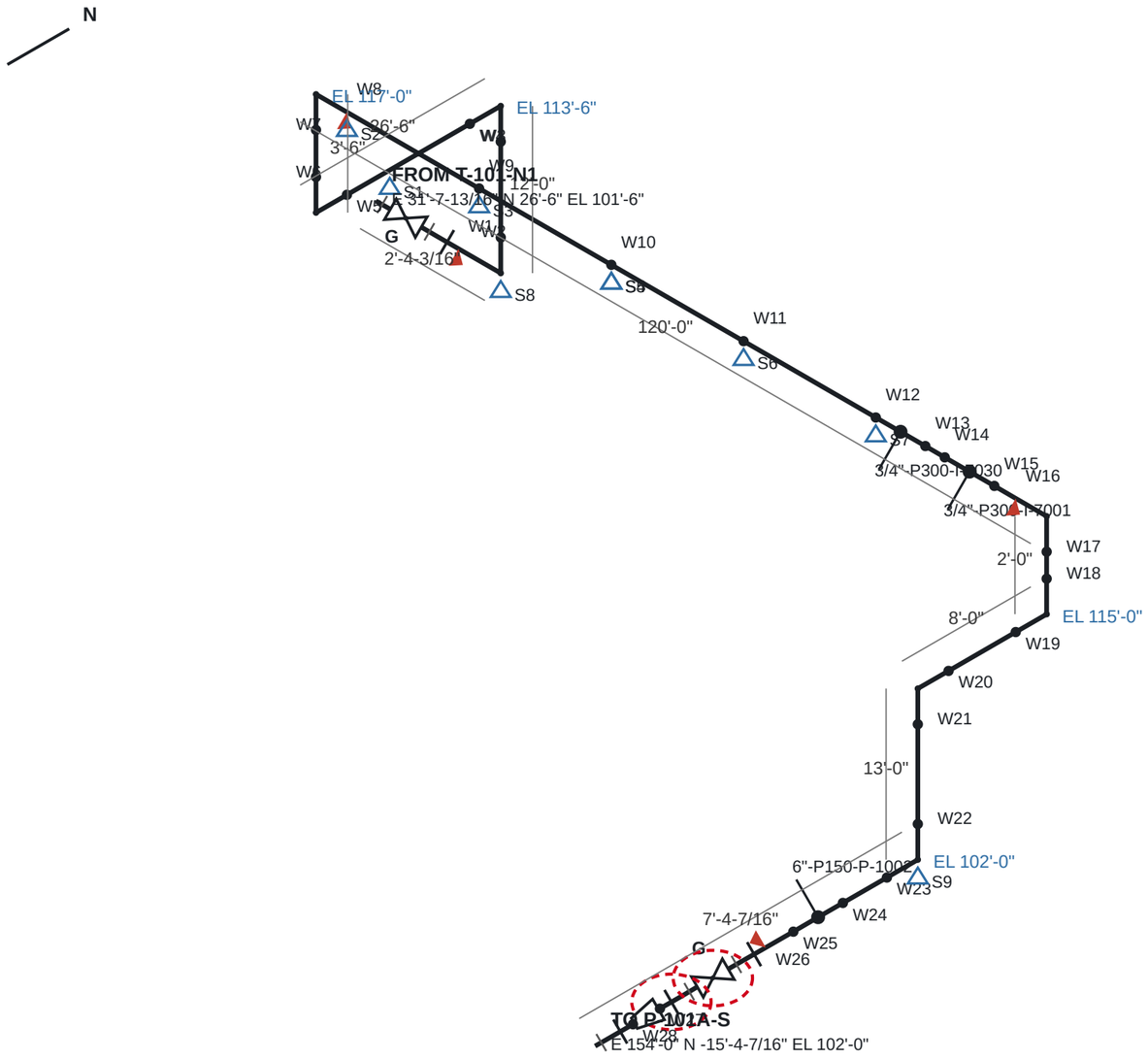
Inspection and test plan (SW-ITP-1..3)

H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

#	Activity	Reference	Acceptance	Record	Contr.	Owner	3rd party	Status	Signed (contr. / owner / 3rd)	Date	Remark
1	Work permit issued; isolation list (valves locked, blinds fitted) verified at zero pressure and drained	SW-WO-3, isolation list	Every isolation in place and tagged; zero pressure at the vent	Signed isolation certificate	H	H	—				
2	As-found alignment recorded	Owner alignment procedure	Recorded	Alignment record	W	R	—				
3	Seal and bearings replaced; impeller and wear rings inspected	Owner pump procedure	Clearances within the owner's criteria	Pump overhaul record	S	R	—				
4	Nozzle joints remade with new gaskets; studs to preload	SW-WO-5	Preload as stated	Bolt-up record	W	W	—				
5	Final alignment	Owner alignment procedure	Within the owner's alignment criteria	Alignment record	W	W	—				
6	Run test at duty	SW-TEST-2	No seal leak; temperatures and vibration within owner limits	Run test record	W	H	—				
7	Blinds removed, isolations returned to normal line-up per the isolation list	SW-WO-3	Every blind removed and accounted for	Isolation certificate closed	H	H	—				
8	Records complete; work order handed back to operations	SW-ITP-3	All ITP activities accepted or not applicable	Signed ITP and completion record	H	H	R				

Isometric ISO-6"-P150-P-1001

ISOMETRIC — 6"-P150-P-1001 — Overhaul pump P-101A (mechanical seal and bearings)



BILL OF MATERIALS

1	GASKET 6" P150 COMPRESSED SHEET	4
2	VALVE, GATE, WEDGE, OS&Y 6" P150	2
3	STUD SET 8 × 0.625" × 4" WITH 2 NUTS, FOR 6" P150	4
4	FLANGE WELD-NECK 6" P150 RAISED FACE	3
5	ELBOW 90° R = 5/3 OD 6" P150	8
6	PIPE 6" P150 BEVELLED ENDS	174'-6-1/8"
7	TEE REDUCING 6" × 3/4" P150	2
8	TEE EQUAL 6" P150	1
9	REDUCER CONCENTRIC 6" × 4" P150	1
10	STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P150	1
11	FLANGE WELD-NECK 4" P150 RAISED FACE	1
12	GASKET 4" P150 COMPRESSED SHEET	1

WELDS 28 (4 FIELD) · SUPPORTS 9 · SPOOLS 6

W1F W2 W3 W4 W5 W6 W7 W8F W9 W10
W11 W12 W13 W14 W15 W16F W17 W18 W19 W20
W21 W22 W23 W24 W25 W26F W27 W28

6"-P150-P-1001

CLASS H150 — Hydrocarbon process
SERVICE P · 6" · P150 · DESIGN 150 psi
T-101-N1 → P-101A-S
PIPE 174'-6-1/8" · WEIGHT 3040 lb

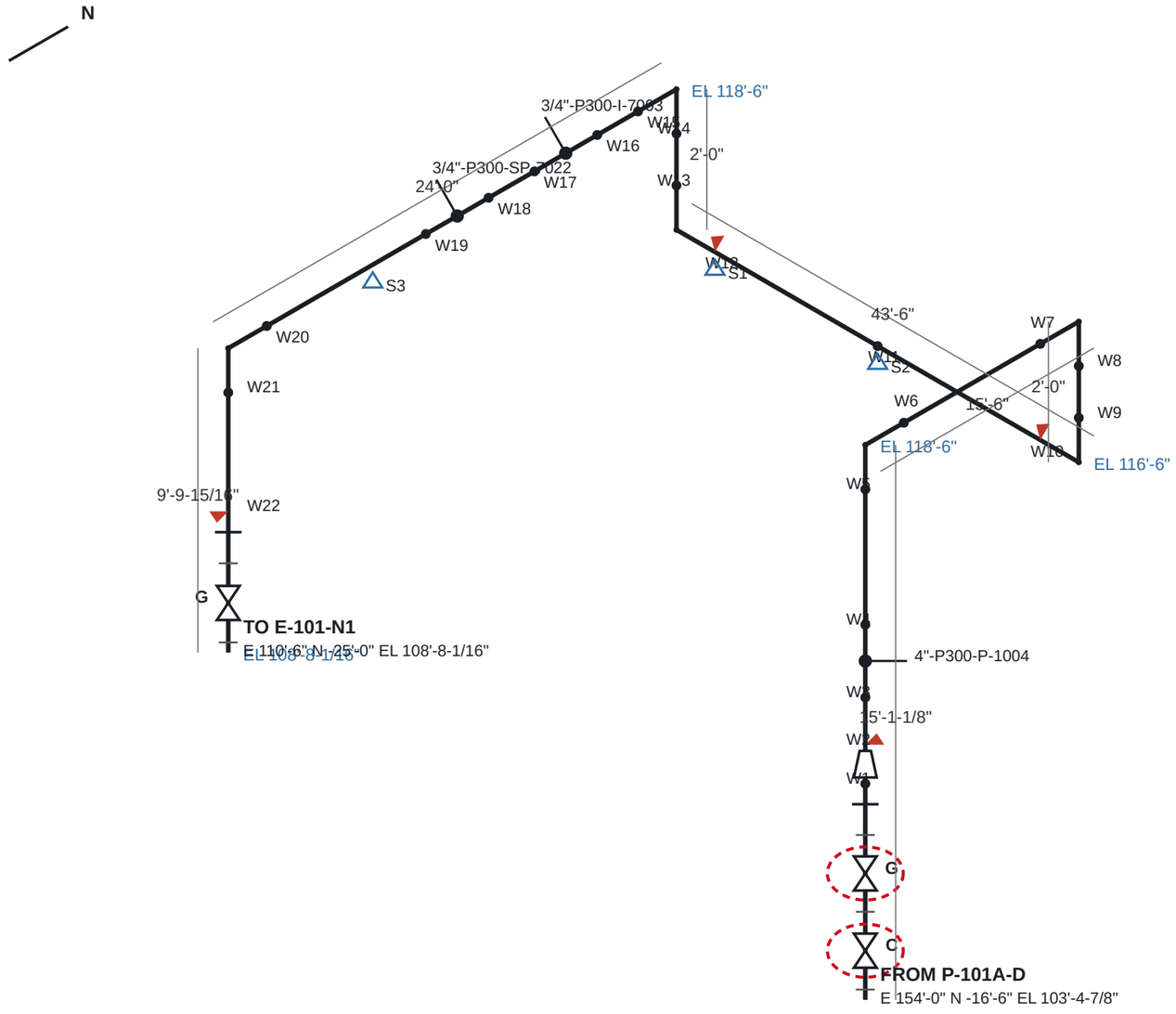
DWG ISO-6"-P150-P-1001 REV 0 WO-400-01

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

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Isometric ISO-4"-P300-P-1003

ISOMETRIC — 4"-P300-P-1003 — Overhaul pump P-101A (mechanical seal and bearings)



BILL OF MATERIALS

1	GASKET 3" P300 COMPRESSED SHEET	3
2	VALVE, SWING CHECK 3" P300	1
3	STUD SET 4 × 0.875" × 5-3/4" WITH 2 NUTS, FOR 3" P300	3
4	VALVE, GATE, WEDGE, OS&Y 3" P300	1
5	FLANGE WELD-NECK 3" P300 RAISED FACE	1
6	REDUCER CONCENTRIC 4" × 3" P300	1
7	PIPE 4" P300 BEVELLED ENDS	98'-8-7/16"
8	TEE EQUAL 4" P300	1
9	ELBOW 90° R = 5/3 OD 4" P300	6
10	TEE REDUCING 4" × 3/4" P300	2
11	STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P300	2
12	FLANGE WELD-NECK 4" P300 RAISED FACE	1
13	GASKET 4" P300 COMPRESSED SHEET	2
14	VALVE, GATE, WEDGE, OS&Y 4" P300	1

WELDS 22 (4 FIELD) · SUPPORTS 3 · SPOOLS 5

W1 W2F W3 W4 W5 W6 W7 W8 W9 W10F
W11 W12F W13 W14 W15 W16 W17 W18 W19 W20
W21 W22F

4"-P300-P-1003

CLASS H300 — Hydrocarbon process
SERVICE P · 4" · P300 · DESIGN 300 psi
P-101A-D → E-101-N1
PIPE 98'-8-7/16" · WEIGHT 1276 lb

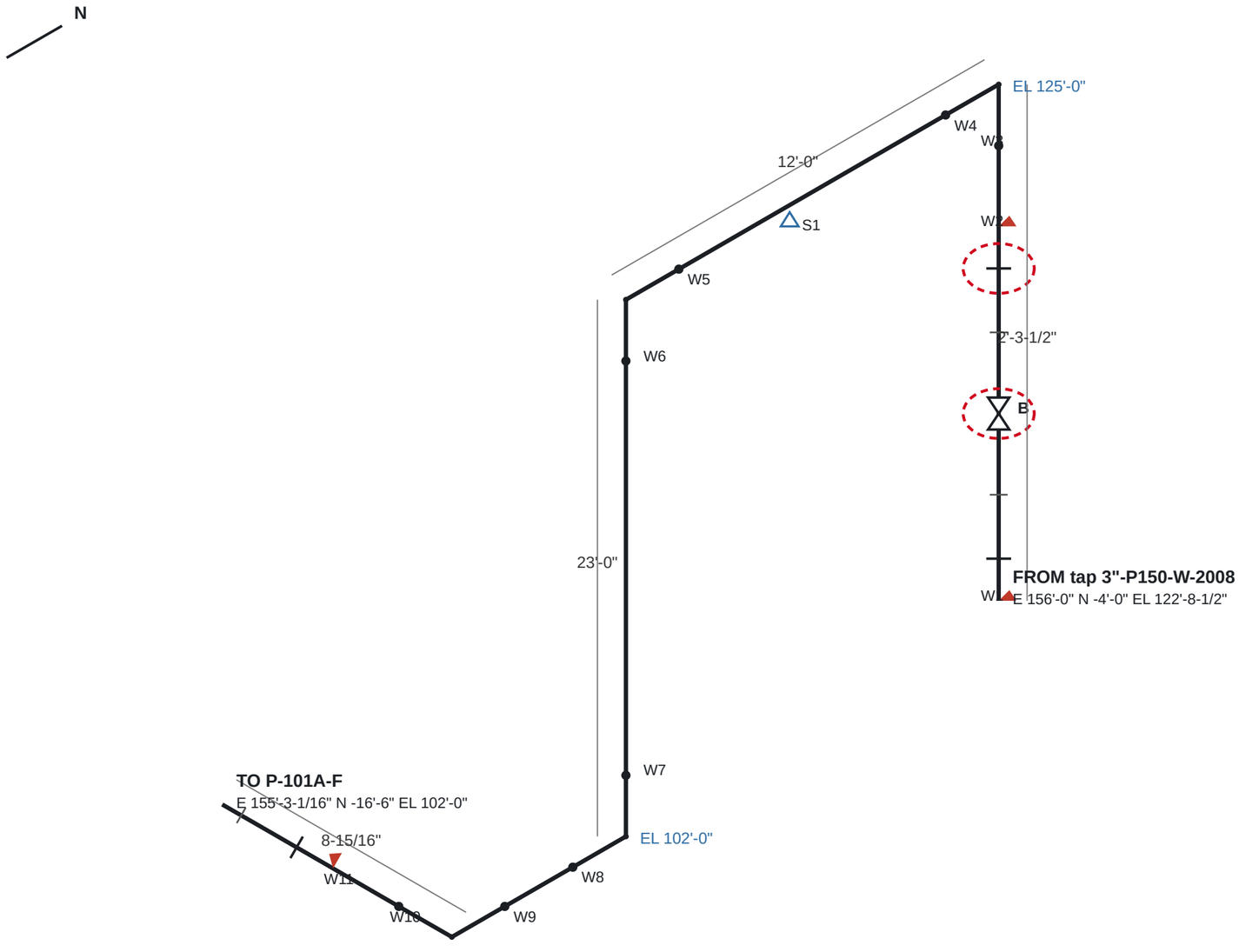
DWG ISO-4"-P300-P-1003 REV 0 WO-400-01

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Isometric ISO-3/4"-P150-W-3017

ISOMETRIC — 3/4"-P150-W-3017 — Overhaul pump P-101A (mechanical seal and bearings)



BILL OF MATERIALS

1	FLANGE WELD-NECK 3/4" P150 RAISED FACE	3
2	GASKET 3/4" P150 COMPRESSED SHEET	3
3	VALVE, BALL, FLOATING 3/4" P150	1
4	STUD SET 4 × 0.5" × 4-1/4" WITH 2 NUTS, FOR 3/4" P150	3
5	PIPE 3/4" P150 BEVELLED ENDS	35'-5-7/16"
6	ELBOW 90° R = 5/3 OD 3/4" P150	4

WELDS 11 (3 FIELD) · SUPPORTS 1 · SPOOLS 4

W1F W2F W3 W4 W5 W6 W7 W8 W9 W10
W11F

3/4"-P150-W-3017

CLASS W150 — Water (cooling and process)
SERVICE W · 0.75" · P150 · DESIGN 150 psi
tap 3"-P150-W-2008 → P-101A-F
PIPE 35'-5-7/16" · WEIGHT 67 lb

DWG ISO-3/4"-P150-W-3017 REV 0 WO-400-01

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